

Research Profile

My research explores haptic interfaces that connect human sensation with digital environments. I investigate how interactions in spatial computing can be translated into meaningful tactile experiences through physical, wearable, and multisensory interfaces.

Research interests: Human-Computer Interaction, Haptics, Virtual Reality, Spatial Computing, Multisensory Interaction

Education

Gwangju Institute of Science and Technology (GIST)

Feb. 2024 - Feb. 2026

M.S. in Intelligent Robotics

Human-Centered Intelligent Systems Lab | GPA: 4.0 / 4.5

Chonnam National University

Mar. 2017 - Feb. 2023

B.S. in Electronics and Computer Engineering

Total GPA: 4.08 / 4.5 | Major GPA: 4.10 / 4.5 | Cum Laude

Research Experience

Research Associate, Human-Centered Intelligent Systems Lab

Mar. 2026 - Present

GIST, Gwangju, Republic of Korea

Graduate Researcher, Human-Centered Intelligent Systems Lab

Feb. 2024 - Feb. 2026

GIST, Gwangju, Republic of Korea

Research Intern, Human-Centered Intelligent Systems Lab

Sep. 2023 - Feb. 2024

GIST, Gwangju, Republic of Korea

Undergraduate Researcher, Visual Information Processing & System Lab

Sep. 2021 - Feb. 2023

Chonnam National University, Gwangju, Republic of Korea

Publications

S. Kang, G. Kim, B. Gim, **J. Park**, J. Um, S. Shin, C. Park, and S. Kim, "When Fingers Become Tools: Rendering Virtual Tool Inertia with a Finger-Mounted Extending Rod." *ACM CHI Conference on Human Factors in Computing Systems*, 2026. [\[Paper\]](#)

S. Kang, G. Kim, B. Gim, **J. Park**, S. Shin, and S. Kim, "EarPressure VR: Ear Canal Pressure Feedback for Enhancing Environmental Presence in Virtual Reality." *ACM Symposium on User Interface Software and Technology (UIST)*, 2025. [\[Paper\]](#)

D. Yeo, G. Kim, M. Oh, **J. Park**, B. Gim, S. Kang, A. Elsharkawy, and S. Kim, "AttrCar: Multisensory In-Car VR with Thermal, Airflow, and Motion Feedback through Built-In Vehicle Systems." *ACM Symposium on User Interface Software and Technology (UIST)*, 2025. **Three Demo Awards** [\[Paper\]](#)

B. Gim, S. Kang, D. Yeo, G. Kim, J. Um, **J. Park**, and S. Kim, "Defying Gravity: Towards Gravitoinertial Retargeting of Acceleration for Virtual Vertical Motion in In-Car VR." *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, 2025. [\[Paper\]](#)

Y. Kang, **J. Park**, S. Hwang, M. Seong, G. Kim, and S. Kim, "You're the One Whom I'm Talking To: The Role of Contextual External Human-Machine Interfaces in Multi-Road User Conflict Scenarios." *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 2025. [\[Paper\]](#)

J. Park, S. Shin, S. Kang, G. Kim, and S. Kim, "Magneto: Enabling Multimodal Haptic Feedback on Paper through Magnetic Fields." *CHI Conference Extended Abstracts*, 2025. [\[Paper\]](#)

S. Hwang, S. Kang, J. Oh, **J. Park**, S. Shin, Y. Luo, J. DelPreto, W. Matusik, D. Rus, and S. Kim, "TelePulse: Enhancing the Teleoperation Experience through Biomechanical Simulation-Based Electrical Muscle Stimulation in Virtual Reality." *ACM CHI Conference on Human Factors in Computing Systems*, 2025. **Best Paper Award (Top 1%)** [\[Paper\]](#)

S. Kang, G. Kim, S. Hwang, **J. Park**, A. Elsharkawy, and S. Kim, "Flip-Pelt: Motor-Driven Peltier Elements for Rapid Thermal Stimulation and Congruent Pressure Feedback in Virtual Reality." *ACM Symposium on User Interface Software and Technology (UIST)*, 2024. [\[Paper\]](#)

S. Hwang, S. Kang, J. Oh, **J. Park**, S. Shin, Y. Luo, J. DelPreto, W. Matusik, D. Rus, and S. Kim, "Proposal of a Framework for Enhancing Teleoperation Experience with Biomechanical Simulation-Based Electrical Muscle Stimulation in Virtual Reality." *UbiComp/ISWC Companion*, 2024. [\[Paper\]](#)

S. Kang, G. Kim, S. Hwang, **J. Park**, A. Elsharkawy, and S. Kim, "Dual-sided Peltier Elements for Rapid Thermal Feedback in Wearables." *IEEE ICRA Workshop on Wearable*, 2024. [\[Paper\]](#)

J. Kim, **J. Park**, J. Jeong, and D. Kim, "Finger Language Translation Device." *IEIE Summer Annual Conference*, 2022. **Outstanding Student Paper Award** [\[Paper\]](#)

Honors & Awards

Best Demonstration Award, IEEE ISMAR Associated work: AttraCar	2025
People's Choice Best Demo Award, ACM UIST Associated work: AttraCar	2025
Demo Honorable Mention Award, ACM UIST Associated work: AttraCar	2025
Best Paper Award (Top 1%), ACM CHI Associated work: TelePulse	2025
Outstanding Student Paper Award, IEIE Associated work: Finger Language Translation Device	Jul. 2022
Full Tuition Scholarship, Chonnam National University	2017 - 2023
Capstone Design Industry-Academic Cooperation Competition Award	Dec. 2021
Oasis Hackathon Special Award	Aug. 2021
Chonnam National University Startup Item Competition	Dec. 2020
Mock Crowdfunding Competition Award	Aug. 2020
Crowdfunding Linked Commercialization Competition Award	Jun. 2020
Student Start-up Promising Team 300	Aug. 2020
Creative Idea Contest Award	Feb. 2020

Technical Skills

Programming: Python, C#, Unity

Prototyping & Fabrication: Arduino, 3D printing, laser cutting, CNC machining

3D Modeling: SolidWorks, Autodesk Inventor, Fusion 360, Blender

Design & Media: Adobe Illustrator, Photoshop, Premiere Pro, Final Cut Pro